

Comparative Hands-On Evaluation of 15 Endpoint Security Products Across 6 Real-World Attack Scenarios

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Evaluated Scenarios

- USB HID Attack
 - Traditional Command and Control
 - Cloud-Based Command and Control
 - Traditional Ransomware
 - IFRET (Indirect File Replacement Encryption Technique)
 - Remote Encryption Workflow
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Key Focus Areas

- Prevention
 - Detection
 - Post-Incident Visibility
 - Defense-in-Depth
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Disclaimer

This research was conducted in an isolated lab environment for defensive security purposes only.

The objective of this study is to evaluate how endpoint security solutions respond to controlled attack behaviors under a maximum available protection configuration. This paper does not aim to rank vendors, promote offensive activity, or provide weaponized tooling.

All testing was performed on systems owned and controlled by the researcher.

Abstract

Endpoint Detection and Response (EDR) solutions are widely used to detect, prevent, and investigate endpoint-based threats. However, practical differences between products are often difficult to measure using marketing claims or feature lists alone.

This study presents a practical hands-on evaluation of 15 endpoint security products against six real-world attack behavior scenarios, including USB HID-based execution, traditional command-and-control communications, cloud-based command-and-control over trusted HTTPS services, ransomware-like file encryption, indirect file replacement encryption (IFRET), and remote encryption workflows targeting network-accessible data.

The objective was to assess how modern endpoint security products respond to realistic attacker behaviors under controlled laboratory conditions, with a focus on prevention, detection, post-incident visibility, and overall defensive effectiveness.

Each product was tested using maximum available protection configurations within an isolated laboratory environment. The results indicate that while most products performed well against traditional ransomware-like behavior, several visibility and prevention gaps were observed in scenarios involving HID emulation, encrypted communication channels, indirect file replacement workflows, and remote encryption over SMB and NAS-based storage paths.

Rather than identifying a universal “best” product, this research highlights practical detection limitations, defensive assumptions, and areas where organizations should validate their EDR deployments beyond baseline testing and conventional attack simulations.

1. Introduction

Modern organizations increasingly rely on Endpoint Detection and Response (EDR) platforms as a primary defense layer against endpoint-based threats. These solutions combine telemetry collection, behavioral analytics, threat detection, automated response, and forensic capabilities to identify malicious activity that traditional antivirus solutions may fail to detect.

Over the last decade, EDR products have significantly improved their ability to detect known malware, ransomware campaigns, command-and-control frameworks, and post-exploitation activity. As a result, organizations often assume that deploying an EDR solution provides sufficient visibility and protection against modern adversaries.

However, attackers continuously adapt their techniques to bypass security controls. Rather than relying solely on known malware signatures, modern attacks increasingly focus on abusing legitimate operating system functionality, trusted communication channels, user interaction, and indirect workflows that may not match traditional detection models.

Most public comparisons of EDR solutions focus on vendor feature matrices, marketing claims, or controlled malware simulations. While valuable, these approaches do not always reflect how products behave when exposed to realistic attacker techniques under operational conditions.

The purpose of this research is to perform a practical hands-on evaluation of multiple EDR products using real attack behaviors executed in a controlled laboratory environment. The study focuses on observing how products respond to different categories of adversary activity, including USB HID emulation, command-and-control communications, ransomware-like behavior, indirect file replacement workflows, and encryption of remote network shares.

Rather than attempting to identify a single "best" product, the objective is to understand how different security solutions behave when confronted with attack techniques that challenge common defensive assumptions. The findings aim to help defenders, security engineers, and decision-makers better understand the strengths and limitations of endpoint-centric detection approaches.

The evaluated scenarios were selected to represent key stages of the modern attack lifecycle, including initial access, execution, command-and-control, data exfiltration, and impact, while maintaining alignment with relevant MITRE ATT&CK tactics and techniques. This mapping was used to ensure that testing reflected realistic adversary behavior rather than isolated malware execution events.

2. Related Work

Several independent organizations regularly evaluate endpoint security products through public testing programs and comparative assessments.

The MITRE ATT&CK Evaluations project measures vendor visibility and detection coverage against adversary emulation scenarios based on publicly documented threat groups and attack techniques. Similarly, organizations such as AV-Comparatives, SE Labs, and other testing bodies perform comparative assessments of endpoint protection platforms using controlled malware samples and attack simulations.

While these evaluations provide valuable insight into product capabilities, many focus primarily on detection coverage, malware execution, or predefined attack chains. Less attention is often given to unconventional attack workflows that intentionally avoid common detection assumptions, such as indirect file replacement techniques, USB HID abuse, or encryption activities targeting remote network storage.

This study complements existing work by focusing on practical behavioral observations collected during hands-on testing. Rather than measuring detection coverage against predefined attack frameworks alone, the research evaluates how products respond to a set of attacker behaviors that were selected specifically to challenge common endpoint protection models.

3. Test Environment

The evaluation was conducted in a controlled laboratory environment designed to minimize external variables and ensure consistent testing conditions across all products.

Virtualization Platform

All endpoint security products were evaluated inside isolated virtual machines running on Oracle VirtualBox. A clean Windows image was prepared and cloned before each product installation to ensure that every solution was tested under identical conditions.

Endpoint Configuration

Operating System:

- Microsoft Windows 11 Pro (64-bit)
- Build 26200

Virtual Machine Configuration:

- 7 Virtual CPUs
- 25 GB RAM
- 80 GB Virtual Disk
- TPM 2.0 Enabled
- Secure Boot Enabled
- EFI Boot Enabled

Network Configuration

Testing was performed inside an isolated research environment. Internet access was enabled only when required for product activation, updates, telemetry synchronization, and cloud-assisted detection mechanisms.

No production systems, external customer environments, or third-party assets were involved during testing.

Test Preparation

For each evaluated product:

1. A clean Windows image was restored.
2. The selected endpoint security product was installed.
3. Product updates and signature updates were allowed to complete.
4. Maximum available protection features and security controls were enabled where applicable.
5. The attack scenarios were executed.

6. Results, alerts, logs, and prevention actions were documented.
7. The virtual machine was discarded and restored before testing the next product.

4. Methodology

The objective of this study was to observe how different endpoint security products respond to attacker behaviors rather than measuring malware signature coverage alone.

Each attack scenario was executed multiple times under comparable conditions to verify consistency of observed outcomes.

Products were evaluated using four outcome categories:

Prevented

The security product actively blocked the attack, terminated the malicious process, prevented execution, or otherwise stopped the objective from being achieved.

Detected Only

The attack objective was successfully completed; however, the security product generated an alert, detection event, investigation record, or user-visible security notification during execution.

Post-Incident Visibility

No actionable alert or prevention event was generated during execution. However, evidence of suspicious activity could be identified through manual investigation of telemetry, logs, or incident records after completion of the attack.

Not Observed

The attack objective was completed successfully and no meaningful prevention, detection, or post-incident visibility was observed during testing.

For the USB HID scenario, certain products generated alerts associated with external USB device activity or USB storage control policies. These alerts did not necessarily indicate recognition or prevention of HID-based keyboard emulation attacks and are discussed separately within the Results and Discussion sections.

For the IFRET scenario, some products recorded suspicious file-access telemetry that became visible only during manual post-incident investigation. These observations did not result in ransomware alerts, prevention actions, automated response, or user-visible detections during execution and were therefore categorized as Post-Incident Visibility rather than active detection.

The study focused on practical operational outcomes rather than internal detection scores, machine-learning confidence values, vendor-specific risk ratings, or marketing claims. The primary objective was to evaluate real-world defensive effectiveness against representative adversary behaviors aligned with multiple stages of the MITRE ATT&CK framework.

5. Attack Scenarios

5.1 USB HID-Based Attack Simulation

This scenario evaluated the ability of endpoint security products to identify and prevent malicious activity delivered through Human Interface Device (HID) emulation.

Commercially available HID emulation platforms, including devices capable of presenting themselves as standard keyboards, were used to execute predefined command sequences on the target system.

The objective was to determine whether products could distinguish between legitimate user keyboard input and automated attacker-controlled HID activity. Particular attention was given to evaluating whether detections were associated with true HID-based attack prevention or with broader USB device control policies.

5.2 Traditional Command-and-Control Communication

This scenario evaluated detection and prevention capabilities against a conventional command-and-control framework operating over HTTP.

The simulated implant periodically communicated with a remote server, received commands, and returned execution results using traditional beaconing behavior commonly observed during post-exploitation operations.

The objective was to assess behavioral detection capabilities against known command-and-control patterns, including beaconing activity, remote task execution, and outbound communications with attacker-controlled infrastructure.

The traditional command-and-control scenario was implemented using a laboratory deployment of the HAVOC command-and-control framework operating over HTTP.

The framework was selected as a representative implementation of commonly observed post-exploitation command-and-control behavior. The objective was not to evaluate a specific framework, but rather to assess endpoint security visibility, detection, and prevention capabilities against traditional command-and-control activity under controlled laboratory conditions.

5.3 Cloud-Based Encrypted Command-and-Control Channels

This scenario evaluated the use of legitimate cloud services as communication channels.

Instead of relying on traditional command-and-control infrastructure, the simulated implant communicated through widely used cloud-based services over encrypted HTTPS connections.

The cloud-based command-and-control scenario utilized custom-developed communication workflows operating through legitimate cloud services, including Telegram-based communication channels and cloud-hosted document platforms.

The objective was to determine whether endpoint security products could identify suspicious process behavior when attacker communications were embedded within trusted services and encrypted traffic commonly encountered in legitimate business environments.

Particular attention was given to evaluating visibility and prevention capabilities when the communication channel itself appeared benign while malicious activity was concealed within otherwise legitimate service usage.

5.4 Traditional Ransomware Simulation

For controlled laboratory testing, the ransomware simulation was based on behavioral characteristics inspired by publicly documented LockBit ransomware operations. The implementation was adapted for defensive research purposes to emulate file discovery, encryption activity, and large-scale file modification behaviors commonly associated with modern ransomware campaigns.

The objective was not to reproduce the original malware, but rather to evaluate how endpoint security products respond to ransomware-like behaviors under controlled conditions.

5.5 Indirect File Replacement Encryption Technique (IFRET)

This scenario evaluated a research-developed encryption workflow designed to challenge assumptions commonly associated with ransomware detection.

Rather than modifying existing files directly, the workflow relied on indirect file replacement operations combined with alternative file handling sequences intended to reduce reliance on behavioral characteristics frequently associated with conventional ransomware activity.

The objective was to assess whether endpoint security products could identify malicious intent when traditional ransomware indicators were absent or significantly reduced.

The workflow was developed specifically for this research and was designed to evaluate defensive visibility and detection capabilities under controlled laboratory conditions rather than to emulate a specific ransomware family.

Unlike traditional ransomware operations that typically encrypt files in place and generate highly recognizable behavioral patterns, IFRET distributed file handling activities across multiple low-risk operations that individually appeared benign from the perspective of the endpoint security product.

A separate technical overview of the IFRET concept was previously published by the author and is referenced in [24]. The purpose of including IFRET in this study was to evaluate how modern endpoint security products respond when malicious outcomes are achieved through unconventional workflows that do not closely resemble established ransomware behavior.

5.6 Remote Encryption over Encrypted Channels

This scenario evaluated a research-developed distributed encryption workflow targeting remote file repositories and network-accessible data.

Rather than performing encryption directly on the monitored endpoint, files were transferred over encrypted HTTPS communication channels to a remote processing server. Encryption operations were performed externally before the encrypted files were returned to the originating system and replaced the original content.

The objective was to determine whether endpoint security products could identify malicious outcomes when the primary encryption activity occurred outside the monitored endpoint and only file transfer, synchronization, and replacement operations were visible locally.

This workflow was designed specifically for research purposes to evaluate the effectiveness of endpoint-centric detection models when critical attack stages are delegated to external infrastructure beyond the visibility of the endpoint security agent.

The scenario was included to assess potential visibility gaps that may arise when attackers distribute malicious operations across multiple systems rather than performing all actions directly on the compromised endpoint.

5.7 Mapping Test Scenarios to MITRE ATT&CK

To improve the practical relevance of the evaluation, each attack scenario was mapped to one or more techniques within the MITRE ATT&CK framework.

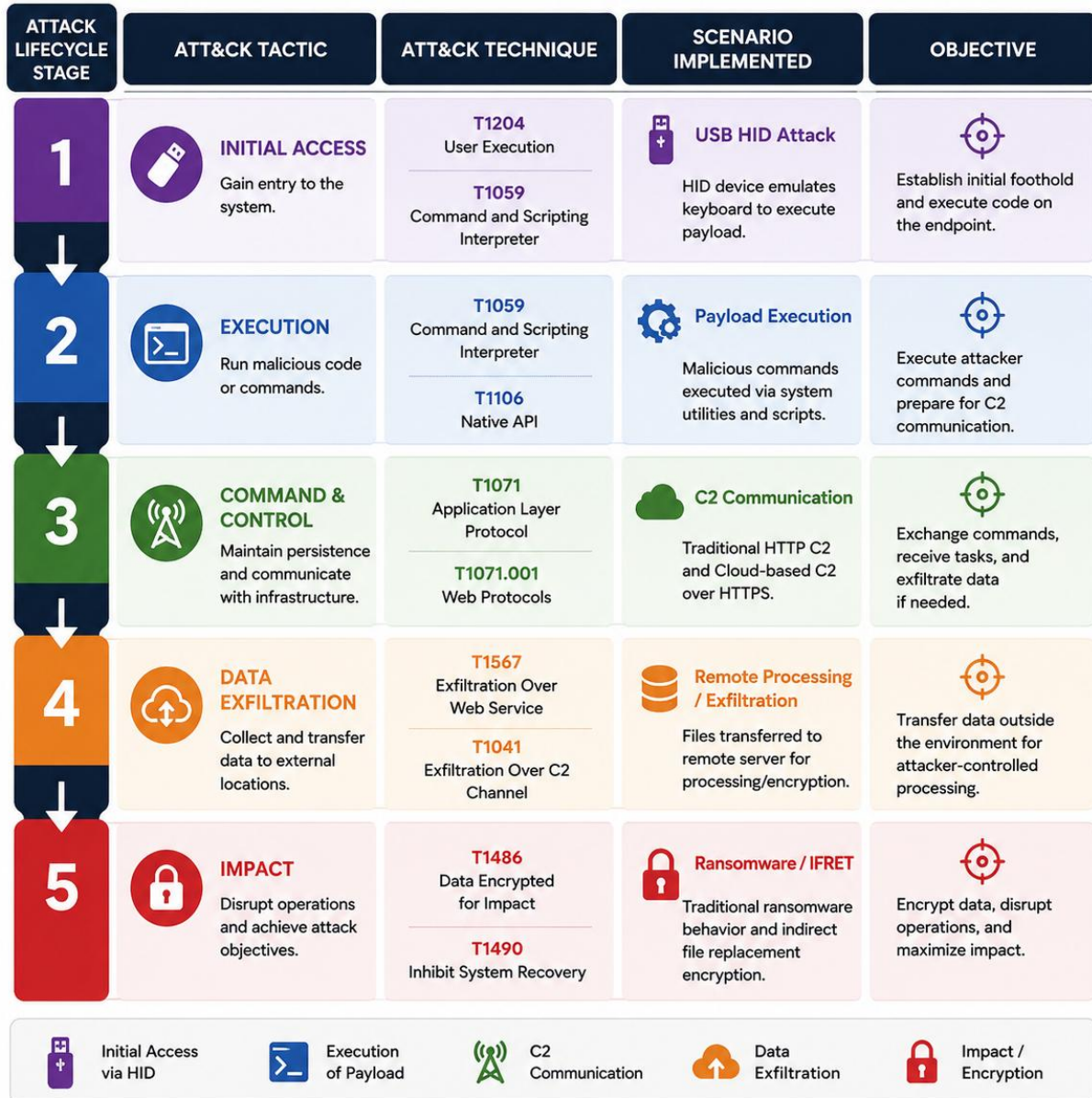
The objective was not to measure ATT&CK coverage scores directly, but rather to assess how endpoint security products responded to representative attacker behaviors commonly observed throughout modern intrusion lifecycles.

Attack Scenario	ATT&CK Tactic	ATT&CK Technique
USB HID Attack	Initial Access / Execution	T1059 Command and Scripting Interpreter, T1204 User Execution
Traditional HTTP C2	Command and Control	T1071 Application Layer Protocol
Cloud-Based C2	Command and Control	T1071.001 Web Protocols
Traditional Ransomware	Impact	T1486 Data Encrypted for Impact
IFRET	Impact / Defense Evasion	T1486 Data Encrypted for Impact
Remote Encryption Workflow	Collection / Exfiltration / Impact	T1567 Exfiltration to Cloud Storage, T1486 Data Encrypted for Impact

The selected scenarios collectively represent multiple stages of the attack lifecycle, including initial access, execution, command and control, data encryption, and data movement operations.

This approach provides a more realistic assessment of defensive visibility than focusing exclusively on malware classification or signature detection.

Figure 1. Attack Lifecycle Coverage Mapped to MITRE ATT&CK



This figure illustrates how the evaluated attack scenarios map to key stages of the MITRE ATT&CK® framework. The scenarios collectively cover the intrusion lifecycle from initial access to final impact.

Figure 2. MITRE ATT&CK Coverage Summary

ATT&CK Tactic	Evaluated
Initial Access	Yes
Execution	Yes
Persistence	No
Privilege Escalation	No
Defense Evasion	Partial
Credential Access	No
Discovery	No
Lateral Movement	No
Collection	Partial
Command and Control	Yes
Exfiltration	Yes
Impact	Yes

This study primarily focused on attacker actions occurring after initial execution and before final impact, with particular emphasis on command-and-control, encryption workflows, and data movement techniques.

6. Results

The evaluation results are summarized in Table 1.

Each product was assessed using the following outcome categories:

- **Prevented:** The attack objective was blocked, interrupted, or otherwise prevented from completing successfully.
- **Detected Only:** The attack objective was successfully completed; however, the product generated an alert, detection event, investigation record, or user-visible security notification during execution.
- **Post-Incident Visibility:** No actionable alert or prevention event was generated during execution. However, evidence of suspicious activity could be identified through manual investigation of telemetry, logs, or incident records after completion of the attack.
- **Not Observed:** The attack objective was successfully completed and no meaningful prevention, detection, or post-incident visibility was observed.

The results presented below reflect behavior observed during testing under maximum available protection configurations within the described laboratory environment.

Table 1. Comparative Results Across All Test Scenarios

Product	USB HID Attack	Traditional C2	Cloud-Based C2	Traditional Ransomware	IFRET	Remote Encryption
CrowdStrike Falcon	Detected Only*	Prevented	Post-Incident Visibility	Prevented	Not Observed	Post-Incident Visibility
Microsoft Defender for Endpoint	Not Observed	Prevented	Not Observed	Prevented	Not Observed	Not Observed
Elastic Security	Detected Only*	Prevented	Detected Only	Prevented	Not Observed	Not Observed
Trend Micro Vision One	Detected Only*	Detected Only	Not Observed	Prevented	Not Observed	Not Observed
Bitdefender GravityZone	Detected Only*	Prevented	Post-Incident Visibility	Prevented	Post-Incident Visibility	Post-Incident Visibility
Kaspersky Endpoint Security	Prevented	Prevented	Post-Incident Visibility	Prevented	Not Observed	Not Observed
Malwarebytes Nebula	Detected Only*	Detected Only	Not Observed	Prevented	Not Observed	Not Observed
FortiEDR	Detected Only*	Detected Only	Post-Incident Visibility	Prevented	Post-Incident Visibility	Post-Incident Visibility
Comodo Advanced EDR	Not Observed	Detected Only	Post-Incident Visibility	Prevented	Not Observed	Not Observed
Acronis Cyber Protect	Not Observed	Detected Only	Not Observed	Not Observed	Not Observed	Not Observed
HP Wolf Security	Detected Only*	Prevented	Not Observed	Prevented	Not Observed	Not Observed
Xcitium OpenEDR	Not Observed	Prevented	Post-Incident Visibility	Prevented	Not Observed	Not Observed
Sophos Intercept X	Detected Only*	Prevented	Post-Incident Visibility	Prevented	Not Observed	Not Observed
ESET PROTECT	Detected Only*	Prevented	Not Observed	Prevented	Not Observed	Not Observed
SentinelOne	Detected Only*	Prevented	Detected Only	Prevented	Post-Incident Visibility	Post-Incident Visibility

6.1 Scenario-Specific Observations

USB HID Attack

Products marked as "Detected Only*" generated alerts associated with external USB device activity or USB storage control policies. However, during testing the evaluated devices operated as Human Interface Devices (HID) and presented themselves to the operating system as standard keyboards rather than storage devices.

In these cases, the alert did not prevent execution of the HID-delivered payload. The automated keystroke sequence continued to execute successfully despite the generated notification.

Consequently, the recorded result reflects alert generation without effective prevention of the HID-based attack workflow.

Only Kaspersky Endpoint Security successfully prevented payload execution by blocking HID interaction rather than relying solely on USB storage control policies.

IFRET

During the IFRET scenario, several products recorded telemetry associated with unusual file access patterns, elevated file interaction rates, or suspicious modification activity.

However, these observations did not result in ransomware alerts, prevention actions, automated response, or user-visible detections during execution.

In some cases, evidence of suspicious behavior was only observable through manual post-incident investigation after the attack had already completed.

As a result, affected files were successfully processed and the attack objective was achieved without interruption.

For the purpose of this study, successful post-incident telemetry collection alone was not considered equivalent to detection or prevention because no actionable alert was generated during execution and no protective action was taken before completion of the attack workflow.

Table 2. Summary of Outcomes by Attack Scenario

Attack Scenario	Prevented	Detected Only	Post-Incident Visibility	Not Observed
USB HID Attack	1	9	0	5
Traditional C2	8	5	0	2
Cloud-Based C2	0	9	0	6
Traditional Ransomware	13	0	0	2
IFRET	0	0	3	12
Remote Encryption	0	0	3	12

Table 3. Overall Outcome Distribution

Outcome Category	Total Observations
Prevented	22
Detected Only	23
Post-Incident Visibility	6
Not Observed	39

6.2 High-Level Observations

Several notable trends emerged during testing.

First, most products demonstrated strong performance against traditional ransomware-style behavior. This outcome was expected because modern ransomware detection has become a mature capability across many endpoint security platforms.

Second, significantly weaker performance was observed in scenarios involving encrypted command-and-control communications through trusted cloud services. While some products generated behavioral detections, no tested product consistently prevented communication across all cloud-based command-and-control scenarios.

Third, USB HID emulation exposed important differences between products. Several solutions generated alerts related to external device activity but did not prevent the simulated HID-based attack sequence from executing successfully.

Finally, the indirect file replacement encryption workflow and remote encryption scenarios produced the lowest detection and prevention rates across all evaluated products. Under the tested conditions, none of the evaluated products generated prevention actions against these workflows.

7. Discussion

The observed results highlight several important characteristics of modern endpoint security products and reveal differences between protection mechanisms, visibility capabilities, and behavioral assumptions.

7.1 USB HID-Based Attacks

One of the most interesting observations involved USB HID emulation devices.

Several products generated alerts immediately after device connection and classified the activity as external USB device usage. At first glance, this behavior appeared to indicate successful prevention.

However, further investigation revealed that many of these detections were associated with USB storage control policies rather than HID-specific protection mechanisms.

The evaluated HID devices identified themselves as standard Human Interface Devices (HID) and interacted with the operating system as keyboards rather than storage devices.

As a result, although alerts were generated regarding external USB activity, the automated keyboard payloads continued to execute successfully in several cases.

This distinction is important because preventing USB storage devices and preventing malicious HID emulation represent fundamentally different security challenges.

The findings suggest that organizations should not assume that USB storage control policies alone provide protection against HID-based attack platforms.

7.2 Traditional Command-and-Control Activity

Most products demonstrated stronger performance against traditional command-and-control frameworks.

This result is consistent with the maturity of modern behavioral analytics and the extensive industry focus on post-exploitation tooling over the last decade.

Products that successfully prevented execution generally identified suspicious process behavior, command execution patterns, or known command-and-control indicators before the attack objective was achieved.

7.3 Cloud-Based Encrypted Command-and-Control Channels

Encrypted communications over legitimate cloud services proved significantly more difficult to address.

Unlike traditional command-and-control infrastructure, cloud services represent trusted destinations that are frequently accessed by legitimate users and business applications.

This creates a visibility challenge for endpoint security products because the communication channel itself appears legitimate while the malicious activity is embedded within otherwise normal encrypted traffic.

Several products generated behavioral detections, but prevention rates remained noticeably lower than those observed in traditional command-and-control scenarios.

7.4 Indirect File Replacement Encryption Technique (IFRET)

The IFRET workflow produced the most consistent bypass results during testing.

Unlike traditional ransomware behavior, the technique avoided several characteristics commonly associated with ransomware detection models, including direct modification of existing files, immediate file deletion operations, and large-scale extension replacement activities.

Instead, the workflow relied on indirect file handling operations that individually appeared benign from the perspective of the endpoint security solution.

The results suggest that many detection models remain heavily optimized for known ransomware behaviors and may provide reduced visibility when encryption workflows are distributed across multiple low-risk operations.

In several products, suspicious telemetry associated with elevated file interaction activity was recorded during execution. However, these observations did not result in ransomware classifications, prevention actions, or user-visible alerts. In such cases, evidence of suspicious behavior became visible only through manual investigation after completion of the attack workflow.

7.5 Remote Encryption Workflows

The remote encryption scenario exposed an architectural limitation shared by many endpoint-focused security products.

In this workflow, encryption operations occurred outside the monitored endpoint, while the endpoint itself primarily performed file transfer activities through encrypted communication channels.

From the perspective of the endpoint agent, much of the observed activity resembled legitimate network communication and file synchronization behavior.

This finding highlights the challenge of relying exclusively on endpoint telemetry when critical attack stages occur on external infrastructure.

7.6 Implications for Defensive Architecture

A broader lesson emerged from all evaluated scenarios.

No individual product demonstrated complete visibility or prevention across every tested attack category.

Some products performed strongly against traditional malware and ransomware activity, while others showed strengths in device control or behavioral monitoring.

These findings reinforce a well-established defensive principle: endpoint protection should not be viewed as a standalone security strategy.

Organizations should instead adopt a defense-in-depth architecture that combines multiple security layers and telemetry sources.

Examples include:

- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)
- Network Detection and Response (NDR)
- Security Information and Event Management (SIEM)
- Identity-focused monitoring solutions
- Network segmentation and access controls
- Email and web security gateways

The results suggest that combining visibility across multiple security domains provides a more resilient defense model than relying exclusively on any single endpoint security product.

8. Limitations

Every security evaluation contains limitations that should be considered when interpreting the results.

First, all products were evaluated using maximum available protection configurations within the scope of the research environment. While significant effort was made to enable available protection capabilities, enterprise deployments may include additional integrations, custom detections, threat intelligence feeds, security orchestration workflows, and environment-specific tuning that could influence product behavior.

Second, testing was conducted within a controlled laboratory environment. Real-world enterprise environments often contain additional telemetry sources, security controls, network monitoring capabilities, and response workflows that may improve overall detection outcomes.

Third, the study focused exclusively on Microsoft Windows endpoints. Linux, macOS, cloud workloads, containers, and mobile platforms were outside the scope of this evaluation.

Fourth, the selected attack scenarios do not represent the full spectrum of adversary behavior. Modern threat actors employ a wide variety of techniques that were not evaluated during this research.

Finally, endpoint security products evolve continuously through cloud analytics, signature updates, behavioral model improvements, and product releases. Consequently, the results presented in this study reflect observed behavior during the testing period and should not be interpreted as permanent product characteristics.

9. Recommendations

Based on the observations collected during testing, several practical recommendations can be made for security teams and defenders.

Validate Security Controls Through Testing

Organizations should regularly validate their security controls using realistic attack simulations rather than relying solely on vendor documentation or product marketing material.

Evaluate HID-Specific Threats

Device control policies designed to block USB storage devices should not automatically be assumed to provide protection against HID emulation attacks. Organizations should verify how their security stack handles keyboard-emulating devices and similar attack platforms.

Monitor Trusted Communication Channels

Legitimate cloud services and encrypted communication channels can be abused by attackers for command-and-control purposes. Behavioral monitoring should be supplemented with network visibility and threat hunting capabilities where possible.

Expand Visibility Beyond Endpoints

Endpoint telemetry alone may not provide sufficient visibility into attacks involving remote infrastructure, encrypted file transfers, or external processing systems. Additional network-level and identity-level monitoring can improve detection coverage.

Adopt a Defense-in-Depth Strategy

No single security control should be expected to prevent every attack scenario. Organizations should combine endpoint protection with complementary technologies such as:

- XDR platforms
- NDR solutions
- SIEM platforms
- Identity protection systems
- Network segmentation
- Email security controls
- Secure web gateways

Defense-in-depth remains one of the most effective approaches for reducing risk against evolving attack techniques.

10. Conclusion

This research presented a practical evaluation of fifteen endpoint security products across six attack scenarios designed to assess prevention and detection capabilities under controlled laboratory conditions.

The results demonstrated strong performance against traditional ransomware-style behavior and, in several cases, effective detection of conventional command-and-control activity. However, reduced visibility was observed in scenarios involving HID emulation, trusted cloud communication channels, indirect file replacement workflows, and remote encryption operations.

A key observation from this study is that modern endpoint protection products remain highly effective against many established attack techniques, yet no evaluated solution demonstrated complete coverage across all tested scenarios.

These findings support the broader security principle that endpoint protection should be viewed as one component of a larger defensive architecture rather than a standalone security solution.

Future research should expand testing to include additional operating systems, cloud-native workloads, identity-centric attack techniques, and larger-scale enterprise deployments in order to further understand the strengths and limitations of modern detection technologies.

11. Future Work

While this study focused on Windows-based endpoint security products and a limited set of attack scenarios, several areas remain worthy of future investigation.

Linux Endpoint Security Evaluation

Many organizations rely heavily on Linux servers and cloud infrastructure. Future testing should evaluate how endpoint security products perform against Linux-specific attack techniques, persistence mechanisms, and post-exploitation workflows.

macOS Security Assessment

As macOS adoption continues to increase within enterprise environments, comparative testing against modern macOS security platforms would provide valuable insight into cross-platform detection capabilities.

Cloud Workload Protection

Future research should evaluate Cloud Workload Protection Platforms (CWPP) and cloud-native security solutions against container escapes, Kubernetes attacks, cloud credential abuse, and cloud-based persistence mechanisms.

Identity-Centric Attacks

Modern threat actors increasingly target identity systems rather than endpoints alone. Future studies should assess detection capabilities against identity abuse techniques including:

- Pass-the-Hash
- Pass-the-Ticket
- Kerberoasting
- Golden Tickets
- Azure AD abuse
- Privilege escalation through identity systems

Expanded MITRE ATT&CK Coverage

Additional testing should map each attack scenario directly to MITRE ATT&CK techniques and evaluate product visibility and prevention coverage across a broader set of adversary behaviors.

Large-Scale Enterprise Validation

The present study was conducted in a controlled laboratory environment. Future research should validate findings within larger enterprise environments containing multiple telemetry sources, network monitoring systems, identity platforms, and security orchestration capabilities.

The evolving threat landscape requires continuous testing and validation. Future studies should therefore focus on emerging attacker methodologies rather than exclusively on historically documented attack patterns.

12. Acknowledgments

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This work was conducted by the author without vendor sponsorship, commercial influence, or participation from evaluated vendors. All findings, observations, and conclusions presented in this paper reflect the behavior observed during testing under the described laboratory conditions.

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14. Appendix

Appendix A – Evaluated Products

The following endpoint security products were evaluated during the course of this research. Each product was installed, updated, configured using maximum available protection settings where applicable, and tested under identical laboratory conditions.

1. CrowdStrike Falcon
2. Microsoft Defender for Endpoint
3. Elastic Security
4. Trend Micro Vision One
5. Bitdefender GravityZone
6. Kaspersky Endpoint Security
7. Malwarebytes Nebula
8. FortiEDR
9. Comodo Advanced EDR
10. Acronis Cyber Protect
11. HP Wolf Security
12. Xcitium OpenEDR
13. Sophos Intercept X
14. ESET PROTECT
15. SentinelOne

All products were evaluated on Microsoft Windows 11 Pro (Build 26200) within the isolated laboratory environment described in Section 3. Product installation, updates, policy configuration, attack execution, and result collection were performed individually for each evaluated solution to ensure consistency and minimize cross-test contamination.

Appendix B – Attack Scenario Mapping

Scenario	Description
USB HID Attack	HID-based keyboard emulation using commercial HID platforms
Traditional C2	HAVOC-based HTTP command-and-control implementation
Cloud-Based C2	Research-developed Telegram and cloud-service communication workflow
Traditional Ransomware	LockBit-inspired ransomware simulation
IFRET	Indirect File Replacement Encryption Technique (research-developed)
Remote Encryption	Research-developed remote encryption workflow using external processing infrastructure

Appendix C – Product Versions and Test Metadata

Product	Version Tested	Test Date	Configuration
CrowdStrike Falcon	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Microsoft Defender for Endpoint	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Elastic Security	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Trend Micro Vision One	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Bitdefender GravityZone	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Kaspersky Endpoint Security	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Malwarebytes Nebula	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
FortiEDR	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Comodo Advanced EDR	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Acronis Cyber Protect	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
HP Wolf Security	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
Xcitem OpenEDR	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration

Product	Version Tested	Test Date	Configuration
Sophos Intercept X	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
ESET PROTECT	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration
SentinelOne	Latest Available Release Accessible During Testing Period	June 2026	Maximum Available Protection Configuration

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Appendix D – Research Ethics Statement

All testing was performed within systems owned and controlled by the researcher.

No third-party systems, customer environments, production networks, or unauthorized assets were involved during testing.

The objective of the research was to evaluate defensive technologies and improve understanding of endpoint security limitations under controlled conditions.